White hake (*Urophycis tenuis*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.4	2.27	
Complexity in Reproductive Strategy	2.2	2.08	
Dispersal of Early Life Stages	2.32	2.1	
Egg and Larval Survival and Settlement Requirements	2.44	1.4	
Historical Variability in Temperature	1.96	2.4	
Juvenile Habitat Requirements	2.92	2.8	
Other Stressors	1.32	1.45	
Population Growth Rate	2.84	2.36	
Prey Specificity	1.52	2.9	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.16	2.8	
Spawning Cycle	2.84	2.5	
Stock Size/Status	3.04	2.96	
Total Sensitivity | Certainty	**Moderate (2) | 0.59**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.28	1.14	
Bottom Oxygen	2.84	0.96	
Bottom Salinity	2.43	1.11	
Bottom Aragonite Solubility	1.33	0.75	
Small zooplankton integrated biomass	1.26	0.28	
Medium zooplankton integrated biomass	1.37	0.26	
Large zooplankton integrated biomass	1.3	0.34	
Total Exposure	**Moderate (2.01)**	**0.99**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (4.02)**	**1.98**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for White hake. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	Moderate	None
Exposure	High	Moderate	Decrease
Vulnerability	Moderate	Moderate	None
Directional Effect	Negative	Negative	None

Stocks

White hake is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

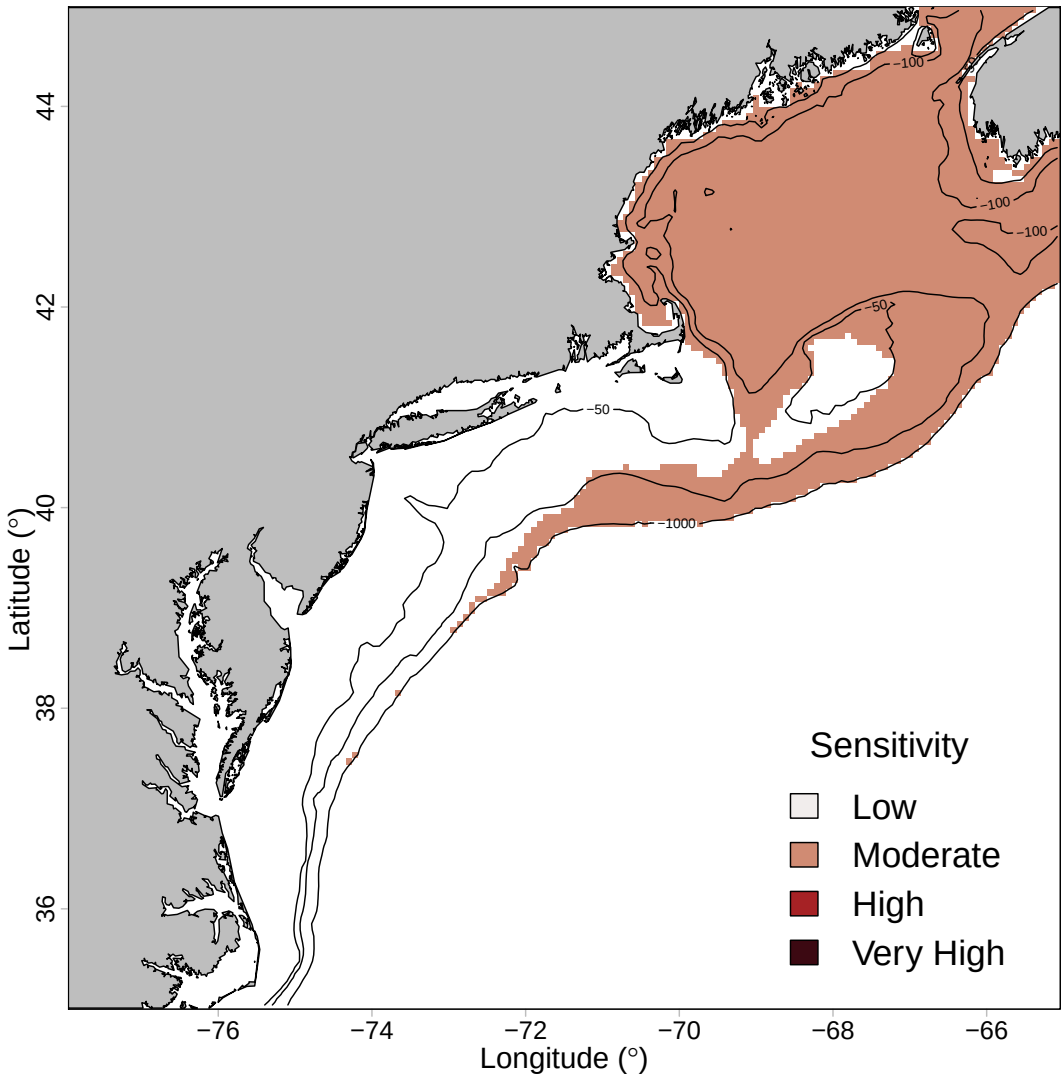


Figure 2: White hake Sensitivity on the Northeast Shelf

White hake received a **Moderate** sensitivity score, with a certainty of 59%. Four attributes had scores greater than 2.5: Stock Size (3), Juvenile Habitat Requirements (2.9), Population Growth Rate (2.8), and Spawning Cycle (2.8) (Figure 1).

Spawning occurs in the spring months along the continental shelf near Georges Bank and the Mid Atlantic Bight (Fahay and Able 1989). Juveniles transition from pelagic to demersal at approximately two months, where eelgrass becomes an important habitat. These habitats are highly vulnerable (Guyant et al. 2022). White hake reach maturity at 2.5-2.8 year and have lifespans of approximately 20 years (Northeast Fisheries Science Center 2013). The stock is under a rebuilding plan, but is not overfished nor is overfishing occurring (Northeast Fisheries Science Center 2025).

White hake also received a Moderate sensitivity score in the 2016 Climate Vulnerability Assessment. This was driven by Stock Status, Population Growth Rate, and Spawning Cycle, similar to our results.

Exposure:

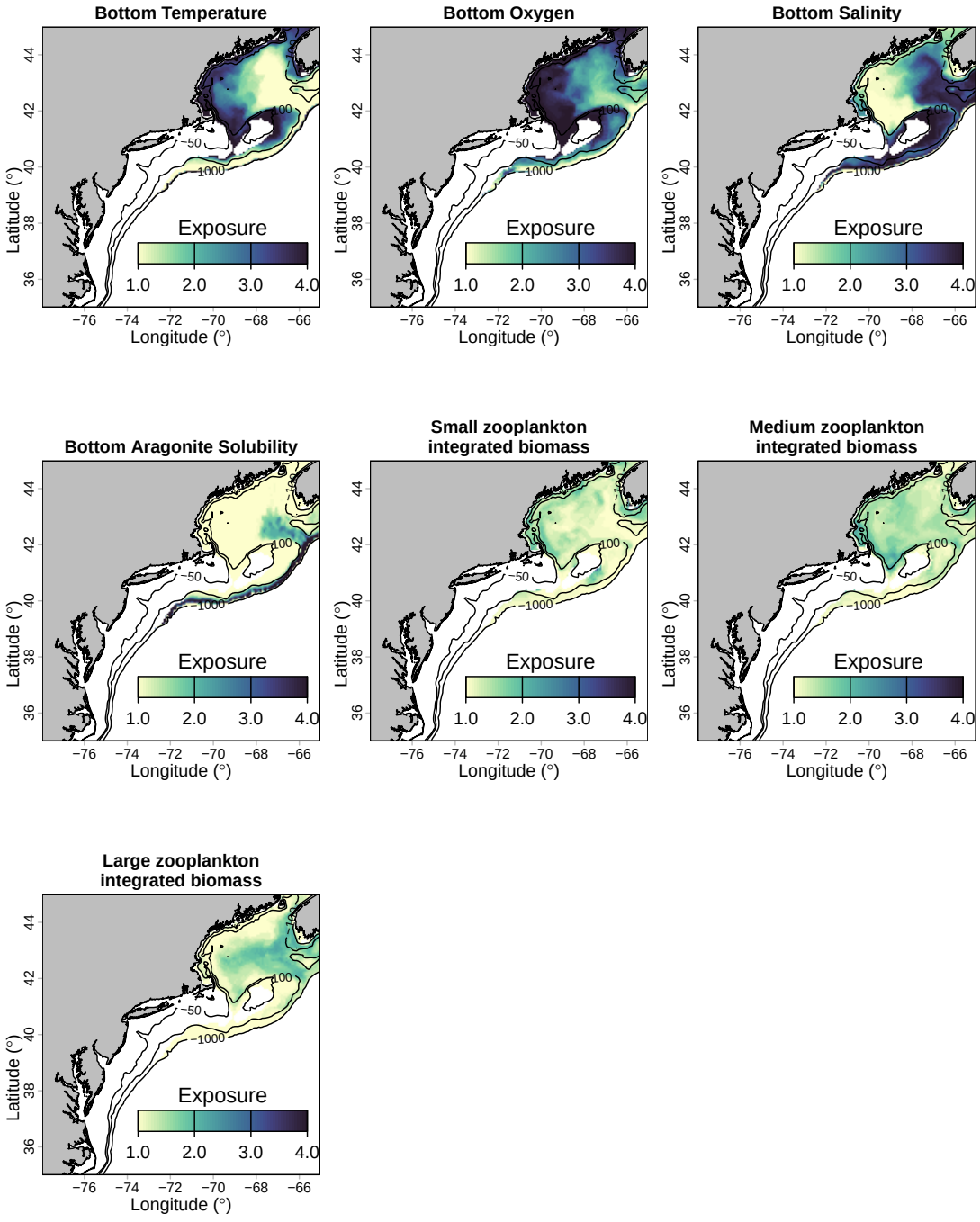


Figure 3: Maps of each Exposure Factor for White hake. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for White hake was **Moderate** (2.01 +/- 0.99) (Figure 1). Exposure was Very High over Georges Bank and Moderate along the coast of Georges Bank. Farther from the coast, scores were Low

(Figure 4). These patterns were driven by bottom temperature, salinity, and oxygen exposure (Figure 3).

In the previous Assessment, White hake received a High exposure score. This previous score was driven by high sea surface temperature and ocean acidification exposure (Hare et al. 2016).

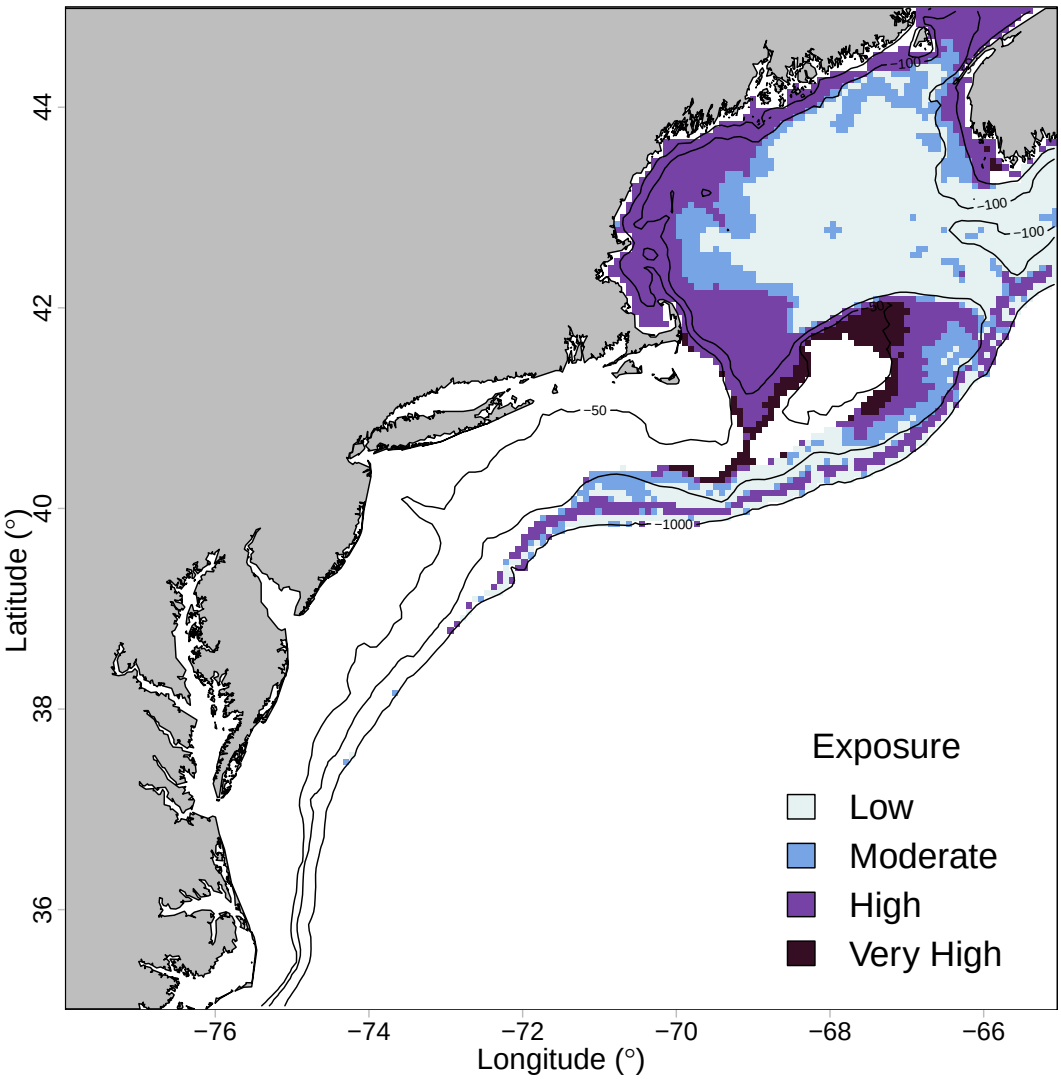


Figure 4: Total Exposure on the Northeast Shelf for White hake. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

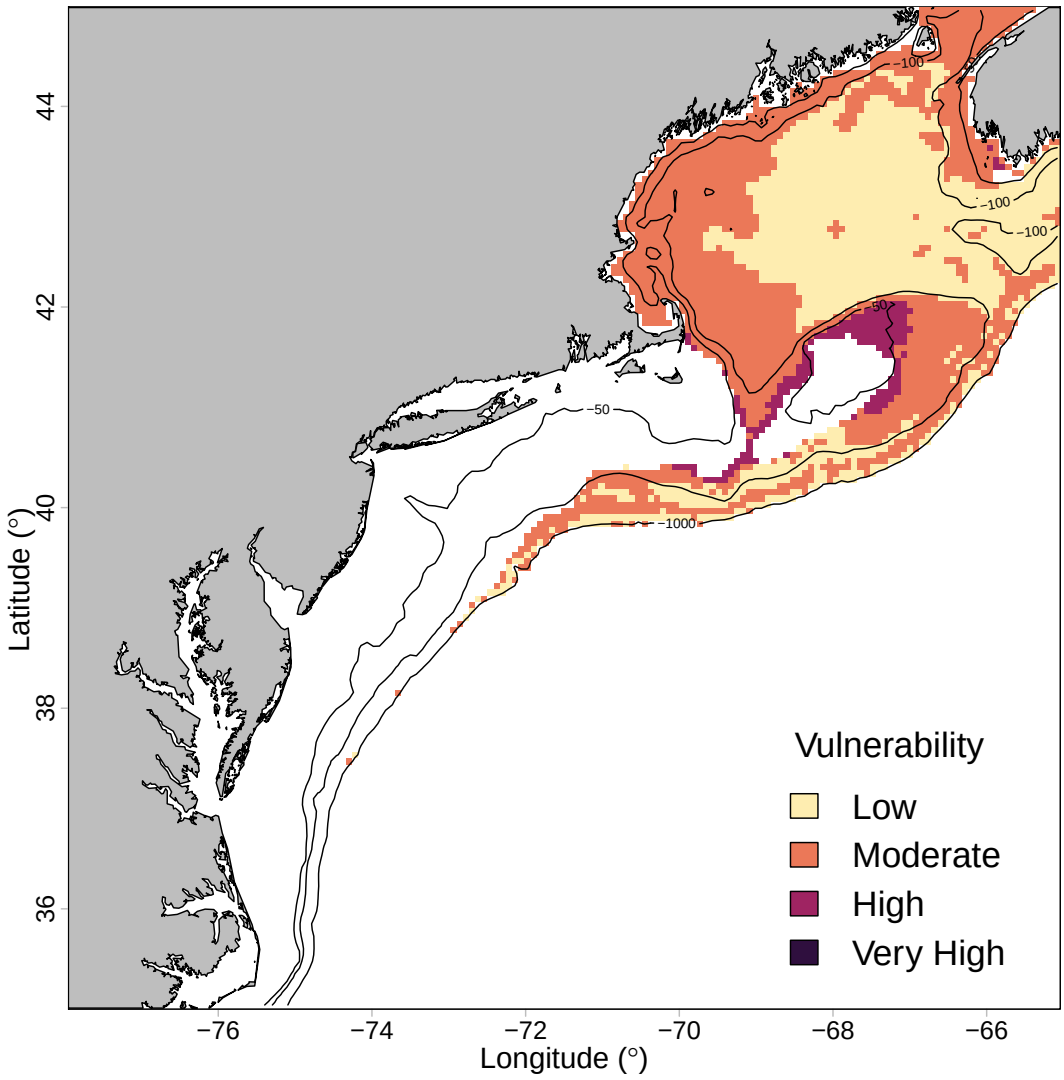


Figure 5: Vulnerability on the Northeast Shelf for White hake. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for white hake was **Moderate** (4.02 +/- 1.98) (Figure 1). Vulnerability was High over Georges Bank and Moderate along the coasts of the Gulf of Maine (Figure 5).

In the previous Assessment, white hake also received a Moderate score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Other Stressors were identified as data gaps (mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The white hake ensemble model received a mean model confidence score of 2.4 (+/- 0.4) indicating moderate confidence in the model. Predicted distributions generally matched expert expectations, however, several noted that seasonal inshore/offshore migrations were missed, possibly due to poor sampling in known canyon habitats. The ensemble model AUC score was 0.89, indicating good model performance.

Additional Metrics

Distribution Shifts:

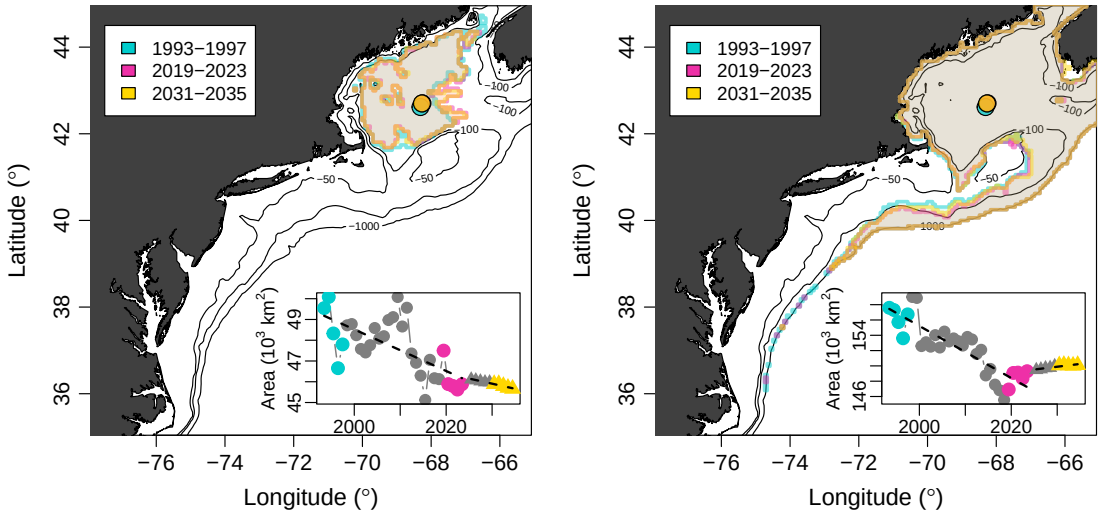


Figure 6: Distribution shifts on the Northeast Shelf for White hake. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

White hake experienced significant declines in both core habitat area and range between 1993 and 2023. They experienced losses primarily on the southern edge of their range. Their range also contracted closer to the continental shelf break (Figure 6). The weighted centroid shifted 11 km north-northeast.

Under near-future ocean conditions, white hake core habitat declines are predicted. However, the overall range is predicted to expand. White hake are predicted to re-occupy areas on the shelf further from the shelf break (Figure 6). The weighted centroid is predicted to shift less than 2 km to the west-southwest.

White hake received a High Distributional Vulnerability rank in the previous assessment (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	48,593	156,517	4.05	4.49
2019-2023	10.7 km NNE	46,152 (-2,441)	148,771 (-7,745)	3.87 (-0.19)	4.25 (-0.24)
2031-2035	1.4 km WSW	45,795 (-356)	150,073 (+1,301)	3.87 (0)	4.33 (+0.08)

Directional Effect:

Experts predicted that white hake will be **negatively** impacted by climate change. Certainty was 70%. This is similar to the results of the previous Assessment (Hare et al. 2016). White hake have a narrow temperature preference (9-13 °C) and as a result, distributions have been shifting northward as bottom waters warm (LaFreniere et al. 2023). Recruitment is strongly linked to circulation patterns, which may be impacted by future change (Han and Kulka 2008) (Ings et al. 2008).

Climate Effects on Abundance and Distribution

White hake are a common groundfish species in the Gulf of Maine, which is warming faster than most of the world's oceans (Pershing et al. 2021). In response to warming in the Gulf of Maine, white hake show a preference for narrow temperature (9-13°C) and depth (55-100 m) ranges and display a gradual northward movement in their center of gravity (LaFreniere et al. 2023). Nye et al. (2009) observed a range contraction in white hake in the Gulf of Maine. White hake may be affected by changes in circulation patterns. Small white hake larvae have been collected in early summer in a Gulf Stream warm core ring seaward of southern Scotian Shelf and southeast of Georges Bank. Patterns of strong recruitment have been linked to the strength of the Labrador Current on the Great Banks and strong onshore transport and downwelling conditions in coastal Newfoundland (Han and Kulka 2008) (Ings et al. 2008).

Life History Synopsis

White hake are a groundfish species inhabiting deep water from Newfoundland to the Mid-Atlantic Bight (Chang et al. 1999). Males reach maturity at 2.5 years and females at 2.8 years; both males and females have a lifespan of around 20 years of age (Northeast Fisheries Science Center 2013). In the Gulf of Maine, white hake of all ages tend to migrate inshore in warmer months and disperse into deeper waters in colder months, while hake in the south move offshore to spawn in the spring; fall movements may be linked to prey movements (Langton et al. 1994). Based on the distribution and abundance of pelagic juveniles, as well as circulation patterns throughout the region, Fahay and Able (1989) suggested that the southern portion of the stock (below the Gulf of St. Lawrence) spawns in early spring (April-May) in deep waters along the continental slope, primarily off southern Georges Bank and the Mid-Atlantic Bight. White hake eggs are buoyant, and hatch in the water column in 3-7 days (Fahay 1983). Larvae may be affected by oceanographic conditions like warm core rings, the Gulf Stream, and Labrador Current for transport (Han and Kulka 2008) (Ings et al. 2008). White hake prefer temperatures between 7-20°C (Chang et al. 1999). Eggs, larvae, and early juveniles are pelagic; older juveniles and adults are demersal. Most catches occur

in Canadian waters at depths between 175 to 375 m with no differences between mature and immature hake. In the Gulf of Maine, depth preferences are 55-100 m (Fahay and Able 1989) (Nozères et al. 2015). Juvenile white hake settle to the bottom at around 2 months of age. Pelagic juveniles are in the upper water column during the daytime. Demersal juveniles inhabit muddy and fine grained sandy bottom sediment (Chang et al. 1999) (Fahay and Able 1989). Demersal juveniles feed primarily on polychaetes, shrimps, and other crustaceans (Bowman 1981). Adults feed on fish, including juveniles of their own and similar species (silver hake), Atlantic herring, Gulf Stream flounder, and possibly alewife, shrimps, krill, and other crustaceans (Langton et al. 1994) (Nozères et al. 2015). The white hake fishery is managed as a single stock by the New England Fisheries Management Council, under the Northeast Multispecies (Small-Mesh) Fishery Management Plan. The stock is under a rebuilding plan, but is not overfished and overfishing is not occurring (Northeast Fisheries Science Center 2025).

Supplemental Scores

Exposure Timeseries

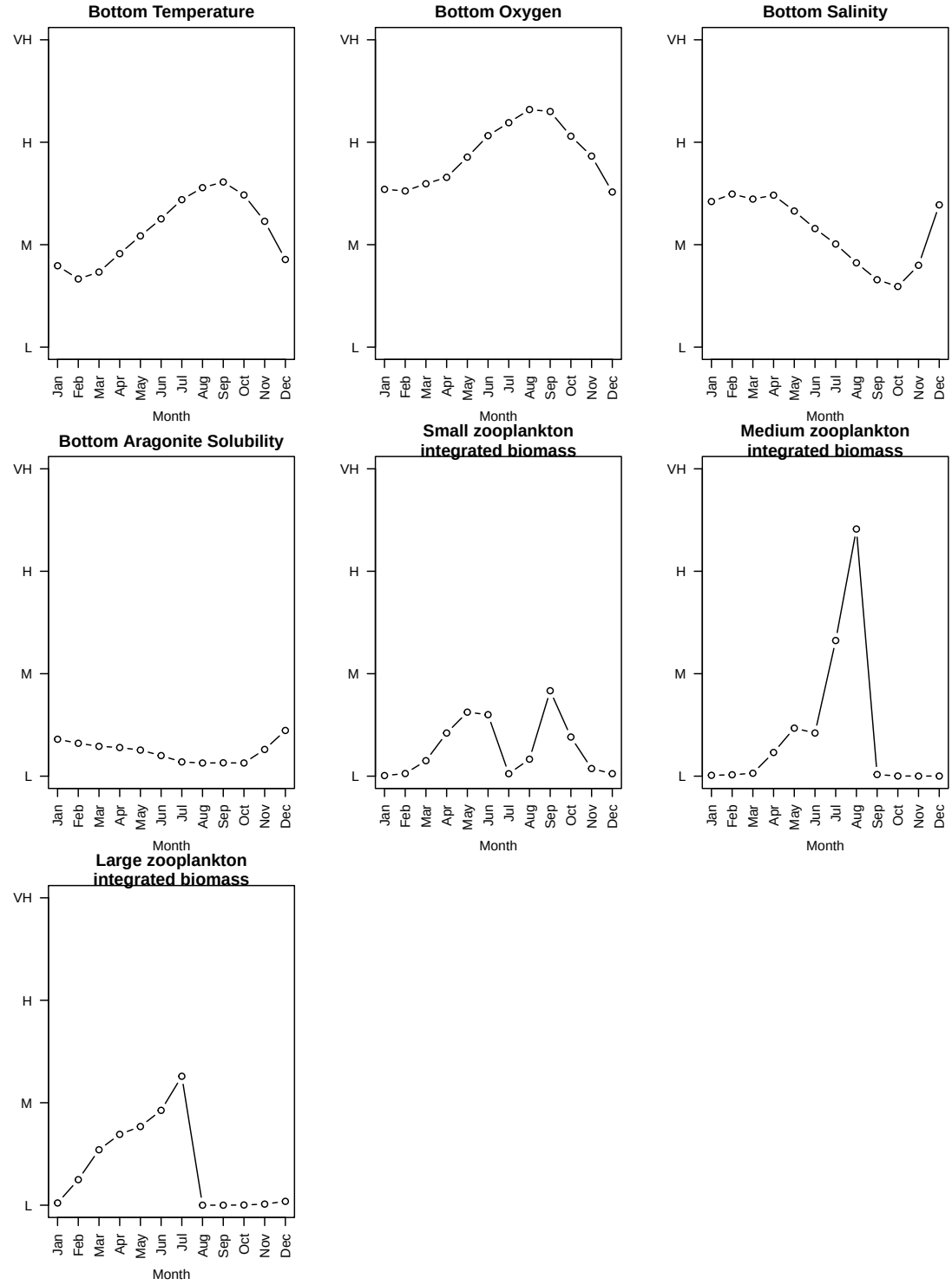


Figure 7: Monthly Timeseries of each Exposure Factor for White hake. Exposure was calculated between 2014-2023 and 2025-2035

Exposure was consistent across months. Exposure was Low for most of the year, and spiked to High in August due to spikes in medium zooplankton biomass exposure (Figure 8; Figure 7).

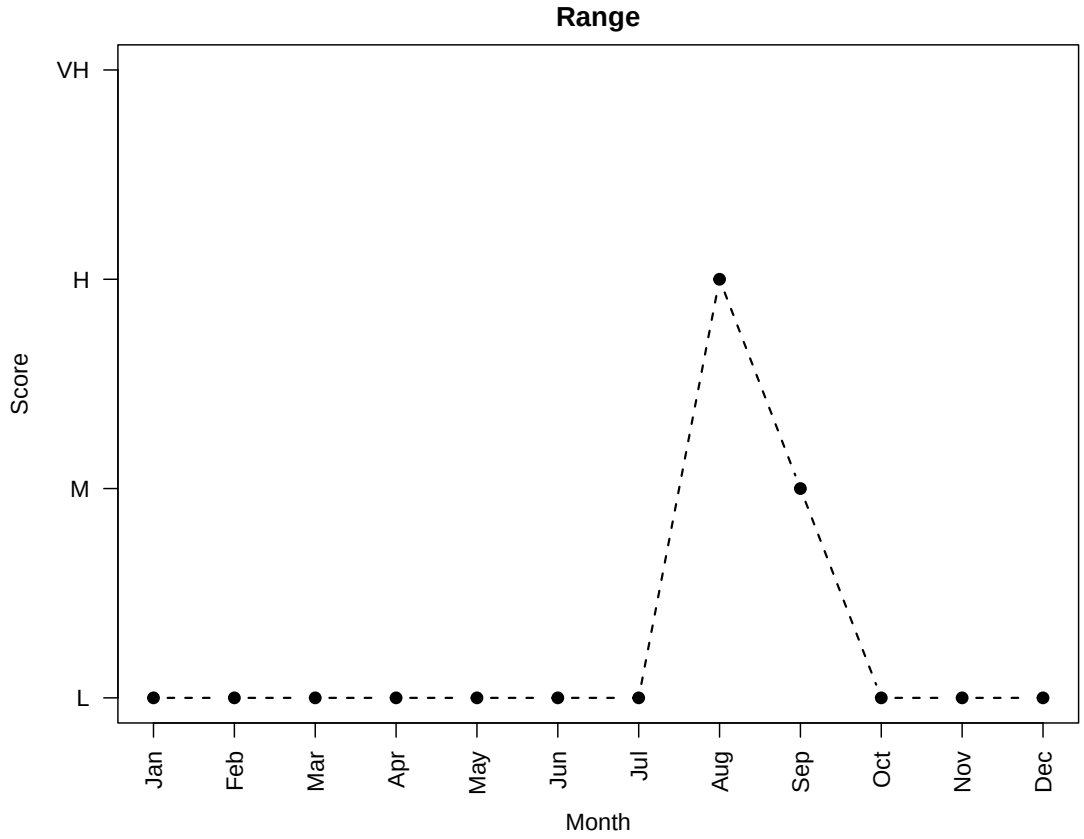


Figure 8: Timeseries of Total Exposure for White hake

Exposure due to Significant Factors

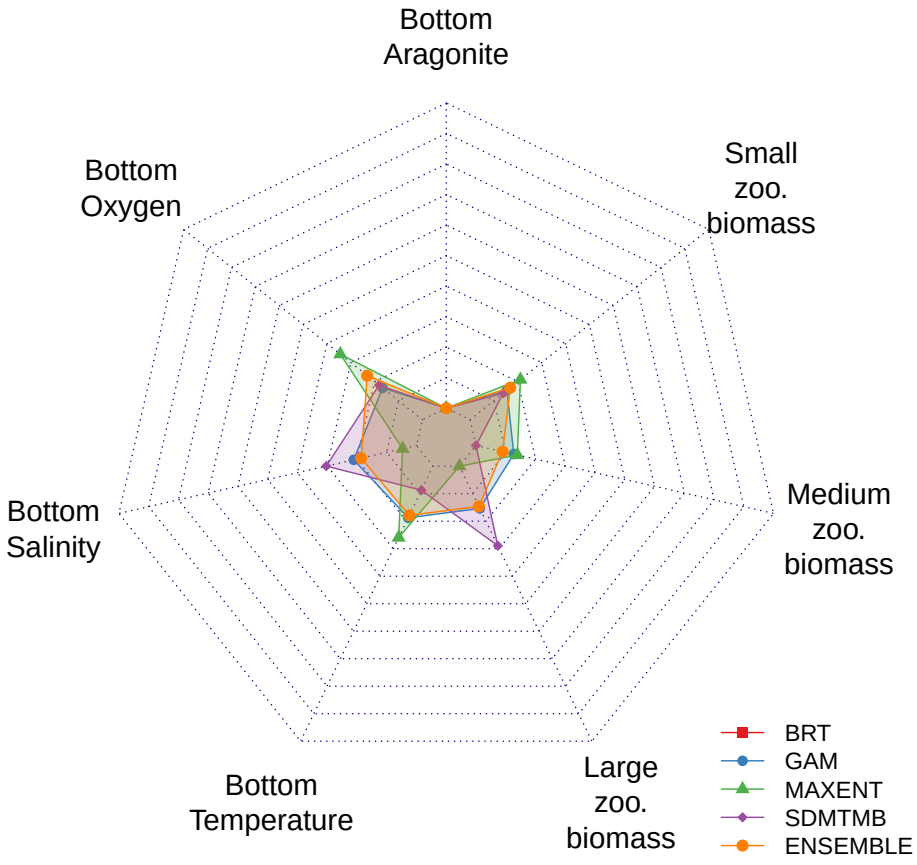


Figure 9: Dynamic Factors driving White hake distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and salinity as well as small and large zooplankton biomass were important dynamic drivers of white hake distribution (Figure 9). White hake are a demersal groundfish species. Their inshore and offshore movements are dictated by temperature and prey movements (Langton et al. 1994). Adults are generalist piscivores (Langton et al. 1994) (Nozères et al. 2015). When only these factors were used to calculate exposure, scores remained Moderate (1.89 +/- 0.99) (Figure 11, Figure 10).

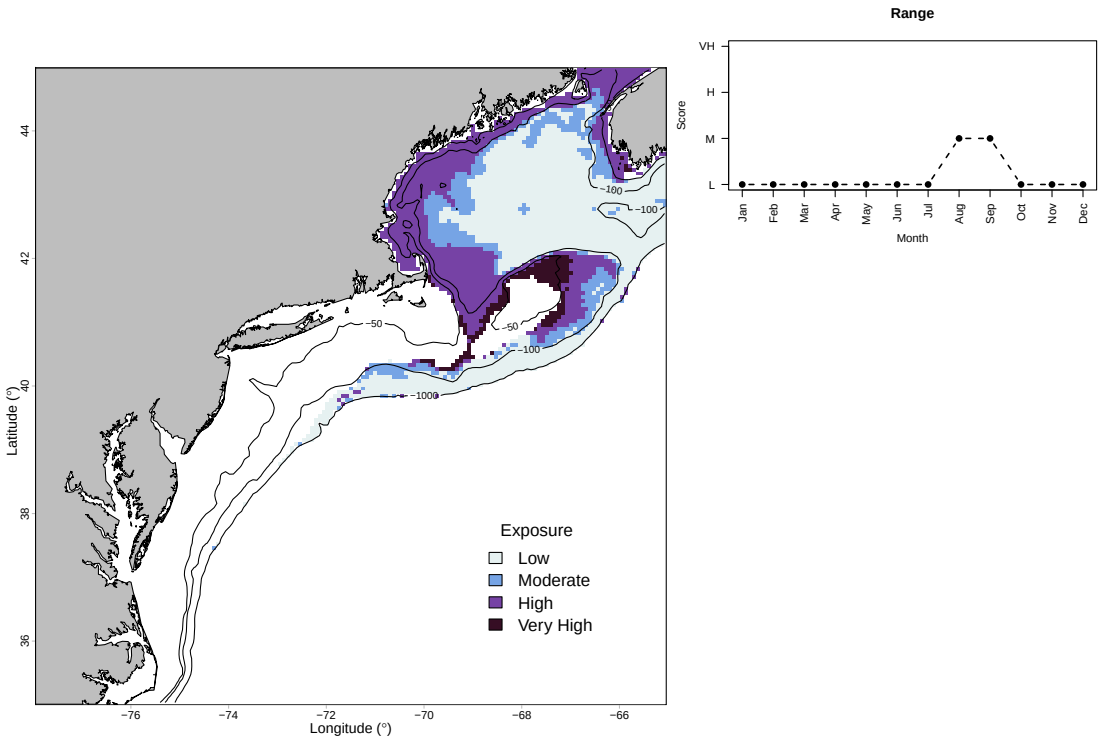
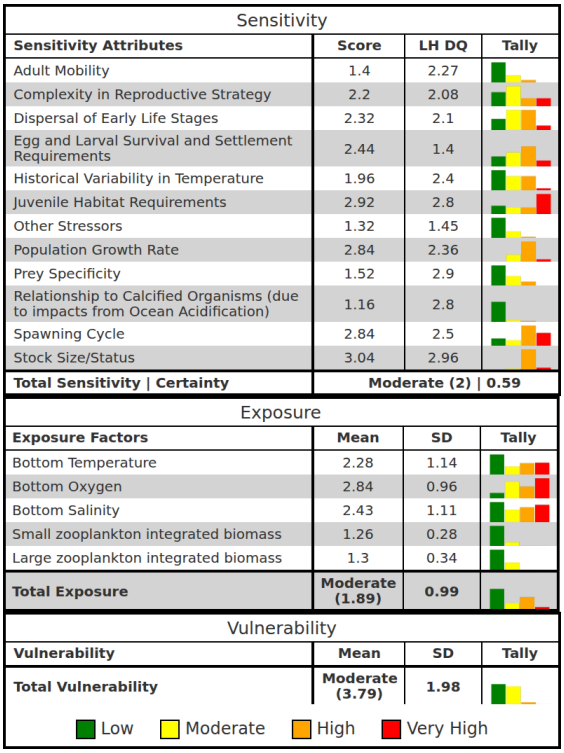


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to White hake distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.4	2.27	
Complexity in Reproductive Strategy	2.2	2.08	
Dispersal of Early Life Stages	2.32	2.1	
Egg and Larval Survival and Settlement Requirements	2.44	1.4	
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Spawning Cycle	2.84	2.5	
Stock Size/Status	3.04	2.96	
Total Sensitivity | Certainty	**Moderate (2) | 0.59**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.28	1.14	
Bottom Oxygen	2.84	0.96	
Bottom Salinity	2.43	1.11	
Small zooplankton integrated biomass	1.26	0.28	
Large zooplankton integrated biomass	1.3	0.34	
Total Exposure	**Moderate (1.89)**	**0.99**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (3.79)**	**1.98**	

Figure 11: As in Figure 1 for White hake, but with only significant Exposure factors and the resulting Vulnerability Scores.

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